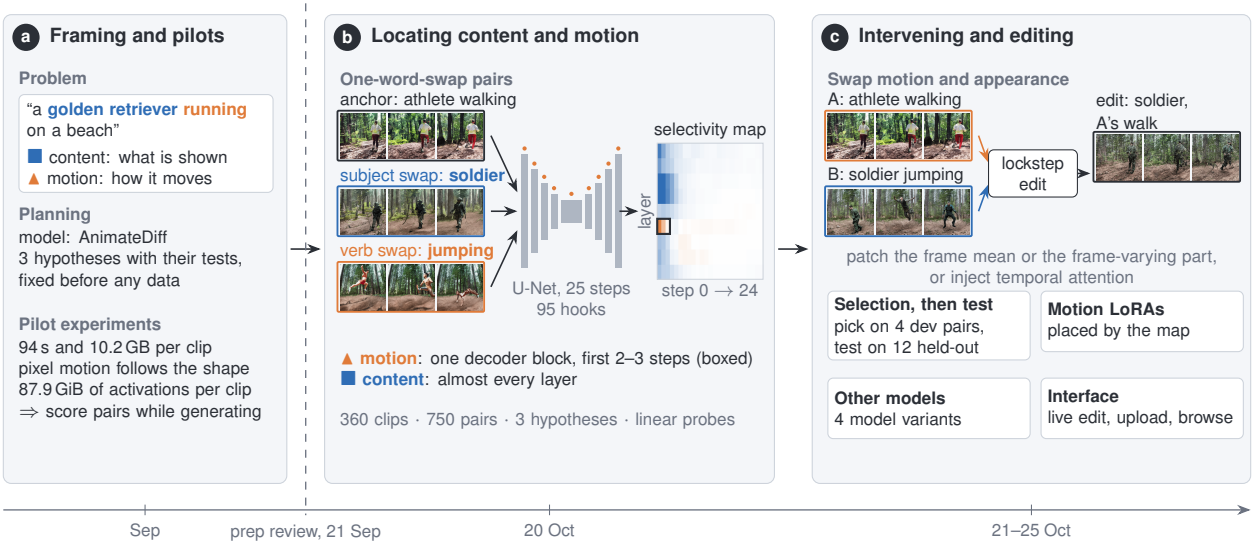

a Framing and pilots
Problem
"a golden retriever running on a beach"
content: what is shown
motion: how it moves
Planning
model: AnimateDiff
3 hypotheses with their tests, fixed before any data
Pilot experiments
94 s and 10.2 GB per clip
pixel motion follows the shape
87.9 GiB of activations per clip
⇒ score pairs while generating
b Locating content and motion
One-word-swap pairs
anchor: athlete walking
subject swap: soldier
verb swap: jumping
U-Net, 25 steps
95 hooks
layer
selectivity map
step 0 → 24
motion: one decoder block, first 2–3 steps (boxed)
content: almost every layer
360 clips · 750 pairs · 3 hypotheses · linear probes
c Intervening and editing
Swap motion and appearance
A: athlete walking
B: soldier jumping
lockstep edit
edit: soldier, A's walk
patch the frame mean or the frame-varying part, or inject temporal attention
Selection, then test
pick on 4 dev pairs, test on 12 held-out
Motion LoRAs
placed by the map
Other models
4 model variants
Interface
live edit, upload, browse
Sep
prep review, 21 Sep
20 Oct
21–25 Oct